BLACK PEN Answer all questions first. **GREEN PEN** Then tick correct answers; rewrite mistakes.

Computational Thinking · ADA

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before looking at the model answers.

1. Name three factors affecting CPU performance.

2. What does ACC store?

3. Distinguish RAM from secondary storage.

4. A processor has more cores but slower clock speed than another. Why can we not predict the fastest from these figures alone?

Today's objectives

Use abstraction to keep the essential details of a problem. Use decomposition to break a problem into manageable parts.

Key vocabulary - Read, cover, recall

abstraction
Keeping the important details of a problem while ignoring irrelevant information.

decomposition
Breaking a complex problem into smaller, manageable parts.

algorithmic thinking
Planning a logical, ordered sequence of steps to solve a problem.

algorithm
A precise sequence of steps or instructions to solve a problem.

Write one definition in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - 2.1.1 / 2.1.2

Abstraction	Focus on the information needed to solve the problem and ignore irrelevant detail. For a journey-time estimate, distance and speed matter; car colour does not.
Decomposition	Break a complex problem into smaller, manageable parts that can be designed and checked separately, such as collecting inputs, calculating a result and displaying it.
Algorithmic thinking	Plan precise, logical steps in the correct order so that the problem can be solved. Decide what happens first, what must be checked and how the answer is produced.
IPO helps organise the steps	Input·Process·Output (IPO) is a useful way to plan the parts of a solution: gather the needed data, process it and show the result. IPO supports decomposition; it is not a replacement for algorithmic thinking.

ABSTRACTION

Keep distance + speed

→

DECOMPOSITION

Input | check | calculate | output

→

ALGORITHM

Read -> validate -> divide -> print

IPO: 60 km, 30 km/h -> 60 / 30 -> 2 hours

IPO supports decomposition; it does not replace ordered steps.

I Do - Worked example

Plan a journey-time calculator using abstraction, decomposition, algorithmic thinking and IPO.

1. ABSTRACTION: retain distance and speed; ignore details such as car colour that do not affect this estimate.
2. DECOMPOSITION: break the task into getting inputs, checking the speed, calculating time and displaying the answer.
3. IPO: input distance = 60 km and speed = 30 km/h; process distance ÷ speed; output journey time in hours.
4. ALGORITHMIC THINKING: in order, collect the values, check speed is greater than zero, divide 60 by 30, then display the result.
5. OUTPUT: 2 hours, assuming constant speed. Each ordered step has a clear purpose and can be checked.

We Do - Guided practice

Use IPO to decompose a lunch-cost calculator into three manageable stages.

Use the model. Write your answer in BLACK PEN:

BLACK PEN Answer all questions first. **GREEN PEN** Then tick or correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

Read the code before you write or test anything.

```
width = 10
height = 8
depth = 4
print(width * height * depth)
```

Predict the raw bitmap size and identify input, process and output.

1. Predict the result in bits.

2. Change width to 20.

3. Identify the three inputs, the processing and one irrelevant detail.

You Do - Independent exam questions

Q1. Define decomposition. [2 marks]

Q2. Explain abstraction in a school timetable application. [3 marks]

Q3. Describe the steps of an algorithm for an overdue-fine calculator using IPO. State two inputs, the process, the output and one way to decompose the task. [5 marks]

Exit Ticket - True or False?

Answer each statement before turning to the answers.

1. Algorithmic thinking means arranging steps in a logical order to solve a problem. ☐ T ☐ F

2. Abstraction means keeping every detail of a problem. ☐ T ☐ F

3. Input-Process-Output can help decompose a problem. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick what is correct, then use the model to fix mistakes.
Do not just copy answers; explain what you changed.

Do Now - Answers

Q1 Clock speed, cache size, core count. ☐

Q2 Results of calculations. ☐

Q3 RAM is volatile working memory; secondary storage is non-volatile. ☐

Q4 Task parallelism, cache and CPU architecture also affect performance. ☐

Code First - Model

10 × 8 × 4 = 320 bits. At width 20 it becomes 640 bits. Inputs are width, height and colour depth; processing multiplies them; output is raw bit count. The image filename is irrelevant to this calculation.

We Do - Model answer

Input: collect quantity and meal price; process: multiply quantity by price; output: display the total cost. Each stage is a smaller part of the problem.

Exit Ticket - Check

1. TRUE - Correct: an algorithm follows ordered, precise steps. Abstraction identifies what matters, decomposition creates smaller tasks, and IPO can organise the inputs, processing and outputs.
2. FALSE - False: keep only relevant information.

3. TRUE - True: IPO organises smaller input, process and output tasks.

You Do - Self-mark your exam answers

Q1 Breaking a complex problem [1] into smaller, manageable problems [1]. ☐

Q2 Retain lesson times and rooms [1]; ignore irrelevant details such as wall colours [1]; this simplifies the problem without removing information needed to produce the timetable [1]. ☐

Q3 Input overdue days [1] and daily fine [1]; then multiply days by the daily fine [1]; finally display the amount due [1]; decompose the program into separate input, calculation and output tasks [1]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW complete answer. Show working: